

Question-2.8.1

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Question

If $(-4, 3)$ and $(4, 3)$ are two vertices of an equilateral triangle. Find the coordinates of the third vertex, given that the origin lies in the interior of the triangle.

Solution

Let

$$\mathbf{A} = \begin{pmatrix} -4 \\ 3 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 4 \\ 3 \end{pmatrix} \quad (1)$$

Consider rotating $\mathbf{A}$ about $\mathbf{B}$ by 60° . Let the rotation matrix be:

$$R_{60^\circ} = \begin{pmatrix} \cos 60^\circ & -\sin 60^\circ \\ \sin 60^\circ & \cos 60^\circ \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} \quad (2)$$

Translate $\mathbf{B}$ to the origin:

$$\mathbf{A}' = \mathbf{A} - \mathbf{B} = \begin{pmatrix} -4 - 4 \\ 3 - 3 \end{pmatrix} = \begin{pmatrix} -8 \\ 0 \end{pmatrix} \quad (3)$$

Solution

Translate back:

$$\mathbf{C} = \mathbf{B} + R_{60^\circ}(\mathbf{A}) \quad (4)$$

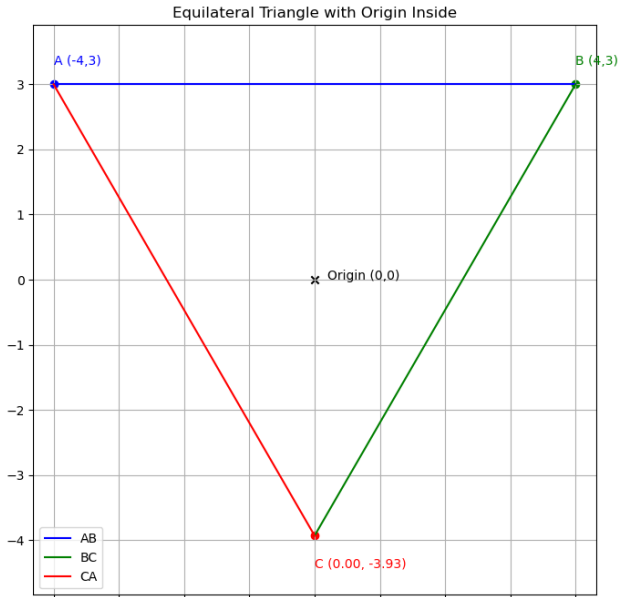
$$= \begin{pmatrix} 4 \\ 3 \end{pmatrix} + \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} -8 \\ 0 \end{pmatrix} \quad (5)$$

$$= \begin{pmatrix} 4 \\ 3 \end{pmatrix} + \begin{pmatrix} \frac{1}{2} \times (-8) + \left(-\frac{\sqrt{3}}{2}\right) \times 0 \\ \frac{\sqrt{3}}{2} \times (-8) + \frac{1}{2} \times 0 \end{pmatrix} \quad (6)$$

$$= \begin{pmatrix} 4 \\ 3 \end{pmatrix} + \begin{pmatrix} -4 \\ -4\sqrt{3} \end{pmatrix} \quad (7)$$

$$= \begin{pmatrix} 4 + (-4) \\ 3 + (-4\sqrt{3}) \end{pmatrix} \quad (8)$$

$$= \begin{pmatrix} 0 \\ 3 - 4\sqrt{3} \end{pmatrix} \quad (9)$$



```
// code.c
#include <stdio.h>
#include <math.h>

void find_third_vertex(double *x, double *y) {
    // Given two vertices
    double Ax = -4, Ay = 3;
    double Bx = 4, By = 3;

    // Side length AB
    double AB = sqrt((Bx - Ax)*(Bx - Ax) + (By - Ay)*(By - Ay));

    // Midpoint M
    double Mx = (Ax + Bx) / 2.0;
    double My = (Ay + By) / 2.0;

    // Height for equilateral triangle
    double h = (sqrt(3) / 2.0) * AB;
```

```
// Two possible third vertices (perpendicular from midpoint)
double C1x = Mx;
double C1y = My + h;

double C2x = Mx;
double C2y = My - h;

// Choose the third vertex such that origin lies inside
triangle
// Since origin is inside, pick vertex with y < 0
if (C1y < 0) {
    *x = C1x;
    *y = C1y;
} else {
    *x = C2x;
    *y = C2y;
}
}
```

```
# Code by GVV Sharma  
# Modified for Problem Solution  
# Released under GNU GPL  
# Calculating area enclosed between curves  
# call.py  
from ctypes import CDLL, c_double, POINTER  
  
# Load the shared library  
lib = CDLL('./libtriangle.so')  
  
# Setup argument types  
lib.find_third_vertex.argtypes = [POINTER(c_double), POINTER(  
    c_double)]
```



```
def find_third_vertex():
    x = c_double()
    y = c_double()
    lib.find_third_vertex(x, y)
    return x.value, y.value

if __name__ == "__main__":
    cx, cy = find_third_vertex()
    print(f"Third vertex coordinates from C code: ({cx}, {cy})")
```

```
import numpy as np
import matplotlib.pyplot as plt
from ctypes import CDLL, c_double, POINTER

# Load the shared library
lib = CDLL('./libtriangle.so')

# Setup argument types for C function
lib.find_third_vertex.argtypes = [POINTER(c_double), POINTER(
    c_double)]

def get_third_vertex():
    x = c_double()
    y = c_double()
    lib.find_third_vertex(x, y)
    return x.value, y.value
```

```
# Given vertices
A = np.array([-4, 3])
B = np.array([4, 3])
C = np.array(get_third_vertex())
Origin = np.array([0, 0])

# Plot the triangle edges
plt.figure(figsize=(7, 7))
plt.plot([A[0], B[0]], [A[1], B[1]], 'b-', label='AB')
plt.plot([B[0], C[0]], [B[1], C[1]], 'g-', label='BC')
plt.plot([C[0], A[0]], [C[1], A[1]], 'r-', label='CA')

# Mark and label points
plt.scatter(*A, color='blue')
plt.text(A[0], A[1]+0.3, "A (-4,3)", color='blue')

plt.scatter(*B, color='green')
plt.text(B[0], B[1]+0.3, "B (4,3)", color='green')
```

```
plt.scatter(*C, color='red')
plt.text(C[0], C[1]-0.5, f"C ({C[0]:.2f}, {C[1]:.2f})", color='
red')
plt.scatter(*Origin, color='black', marker='x')
plt.text(Origin[0]+0.2, Origin[1], 'Origin (0,0)', color='black')

plt.title("Equilateral Triangle with Origin Inside")
plt.legend()
plt.grid(True)
plt.axis('equal')
plt.tight_layout()
plt.show()
```